AL-AQSA UNIVERSITY · BSc ARTIFICIAL INTELLIGENCE ENGINEERING

ENGS 3302 Pattern Recognition

Course Syllabus

First semester 2026/2027 · Version 1, published with Week 2

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Hours	2 hours theory + 2 hours lab per week
Delivery	**Recorded videos every week** are the core of the course. **In-person discussion session** on campus every two weeks for each section, Thursday 12:00–14:00: female section in odd weeks (1, 3, 5, …), male section in even weeks (2, 4, 6, …)
Language	English for all materials and exams
Platform	Moodle: all materials, announcements and submissions. Questions go to Discord or e-mail

About this course

This course teaches how machines recognise patterns: how to turn raw data (images, sounds, measurements) into useful features, how to build classifiers and clustering methods, and how to test whether they work on new data. It follows the development process of a real system: define the problem, develop the concept, choose an algorithm, integrate the system, and test and validate it. Every idea is shown first with a picture and a small numerical example, then with a formula, then in Python code. You are not asked to derive formulas; you are asked to understand them, use them, and compute with them by hand.

Learning outcomes

By the end of the course you will be able to:

Theory

T1 Explain the parts of a pattern recognition system and the kinds of learning.

T2 Represent patterns as feature vectors, compute distances between them, and interpret feature space.

T3 Distinguish a probability from a probability density, apply Bayes' rule to compute posterior probabilities, and find minimum-error and minimum-risk decision boundaries in simple cases.

T4 Analyse how the covariance matrix shapes the decision boundary (linear or quadratic).

T5 Estimate the parameters of a normal distribution by maximum likelihood, and tell parametric methods from non-parametric ones such as k-NN.

T6 Choose suitable evaluation measures and explain overfitting and generalization.

T7 Compare linear classifiers, SVMs, decision trees and ensembles by their assumptions and complexity.

T8 Explain PCA and Fisher's linear discriminant and their role in dimensionality reduction.

T9 Run k-means by hand on small data, tell it apart from hierarchical clustering, and judge the quality of the clusters.

Lab

L1 Use Python, NumPy, pandas and matplotlib to explore, clean, normalise and visualise data.

L2 Program basic classifiers from scratch (nearest mean, Gaussian Bayes, k-NN, Perceptron) and check them against scikit-learn.

L3 Build a complete workflow: preprocessing, training, parameter tuning and cross-validation.

L4 Apply PCA and clustering to real data and interpret the results visually.

L5 Compare several classifiers experimentally and write a technical report that justifies the choice.

L6 Work in a team and present an applied project clearly.

Weekly plan

01	Introduction to pattern recognition: systems, features, the design cycle
02	Features, distances and normalization; training and test data; the nearest-mean classifier
03	Probability for pattern recognition
04	Bayesian decision theory: minimum error and minimum risk
05	Gaussian models: the multivariate normal, maximum-likelihood estimation, LDA, QDA and Naïve Bayes
06	k-nearest neighbours and evaluating classifiers: overfitting, confusion matrix, ROC, cross-validation
07	Linear classifiers: Perceptron, gradient descent, logistic regression
08	Support vector machines
09	Dimensionality reduction and feature selection: PCA, Fisher's discriminant
10	Decision trees and ensembles; trees as readable rules
11	Unsupervised learning: k-means and hierarchical clustering
12	Neural networks (MLP)

The plan may be adjusted slightly during the semester; changes are announced on Moodle.

How each week works

Every week has its own section on Moodle with:

- a one-page **week guide** (what to do, in order, and the deadlines);
- **videos**: about 90 minutes of theory and up to 90 minutes of lab;
- the **Student Notes** (PDF), the study material for the week;
- a **lab notebook** for Google Colab or Jupyter, with self-checks;
- an **assignment notebook** to submit, and sometimes an **interactive page** used in the videos.

Discussion sessions. The in-person sessions support the videos and the Notes: questions, worked problems and discussion. They do not replace the weekly material, so come having watched the videos of the last two weeks.

The exams cover what is in the Student Notes. Nothing outside the Notes is examined, even if it appears in a video, in a book or in a discussion session. The slides support the videos and are not study material.

References

- **Main reference:** R. O. Duda, P. E. Hart and D. G. Stork, *Pattern Classification*, 2nd ed., Wiley, 2001. The Notes give section and page numbers if you want to read more.
- **Supporting, free and legal online:** G. James, D. Witten, T. Hastie, R. Tibshirani and J. Taylor, *An Introduction to Statistical Learning with Applications in Python*, Springer, 2023 (statlearning.com); C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006 (Microsoft Research).

Tools

Python 3 · NumPy · pandas · scikit-learn · matplotlib · Google Colab (Jupyter on your laptop as an offline backup). All tools are free. Lab 1 shows how to set them up.

Assessment

Weekly assignments (from Week 2)	about 10–12%
Other components (exams, final project and its stages)	announced by Week 5

The full grading plan will be published in Version 2 of this syllabus, once the university's exam rules for this semester are known.

Weekly assignments. Each assignment is marked for serious completion, not for every answer being correct:

- **1**: the notebook runs, every task has a real attempt, and the "explain in your own words" questions are answered;
- **0.5**: more than half of the tasks are attempted;
- **0**: not submitted, empty, or copied.

Short oral checks. In the in-person sessions, the instructor asks a few students to explain a part of their own submitted work for about two minutes. If you cannot explain your own work, that assignment receives 0.
Late work. You have 48 hours of grace after a deadline without penalty, for power and internet cuts. Work is not accepted after that. The Week 1 check-in is not graded.

AI tools

You may use AI tools such as chatbots and code assistants, with clear rules for each task. A short guide comes in Week 3. One rule applies from today: **whatever an AI tool gives you, you must be able to explain it and reproduce it yourself.** The oral checks and the exams test exactly that.

Academic integrity

Submit your own work. Discussing ideas with classmates is encouraged; copying code or answers from a classmate, or giving yours to others, is not. Copied work receives 0 for every student involved.

Studying from home

Download each week's Notes and notebooks as soon as you can. They work offline. Keep a fixed weekly rhythm, compute every worked example yourself before reading the answer, and ask early on the course Discord, where one answer helps everyone.